

Online Appendix

Economic Primitives and Classification Prompts

Anthropic Economic Index

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1 Supplementary Figures

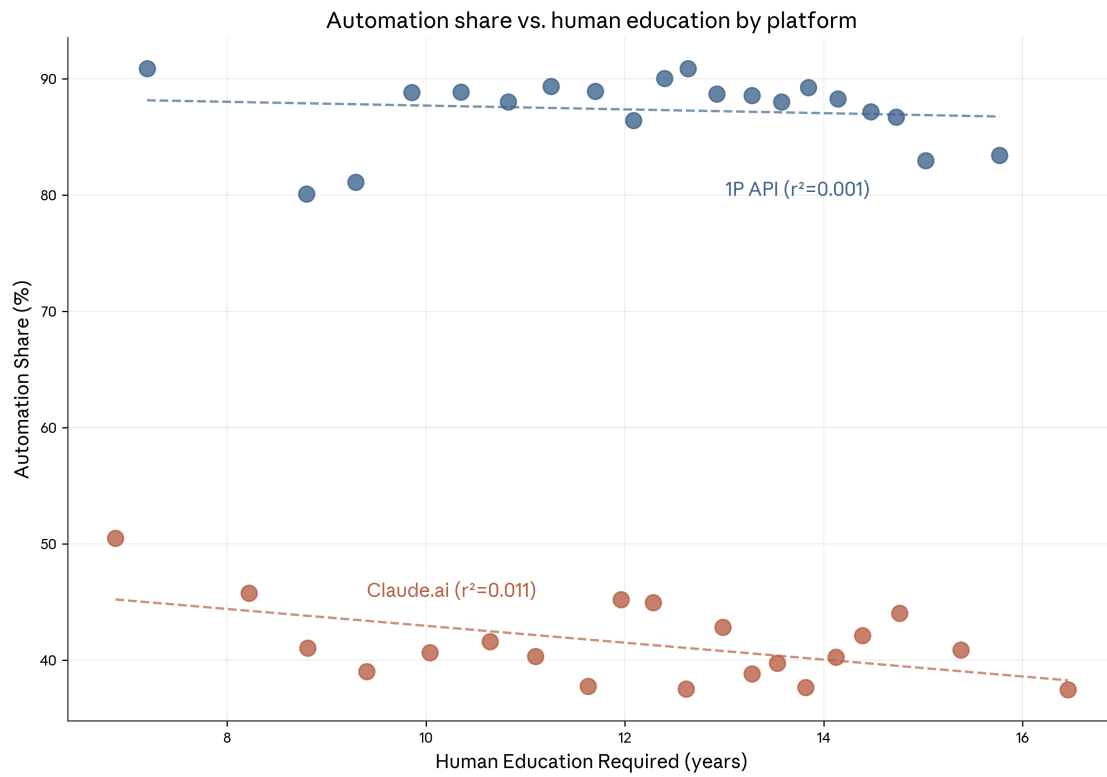


Figure 1: **Automation share vs. Human education years, split by platform.** This figure shows a binned scatterplot of the bivariate relationship between the share of automation and years of formal education needed to understand the User prompts, all measured at the O*NET task level and split by platform. The dashed lines show the fit from a linear regression.

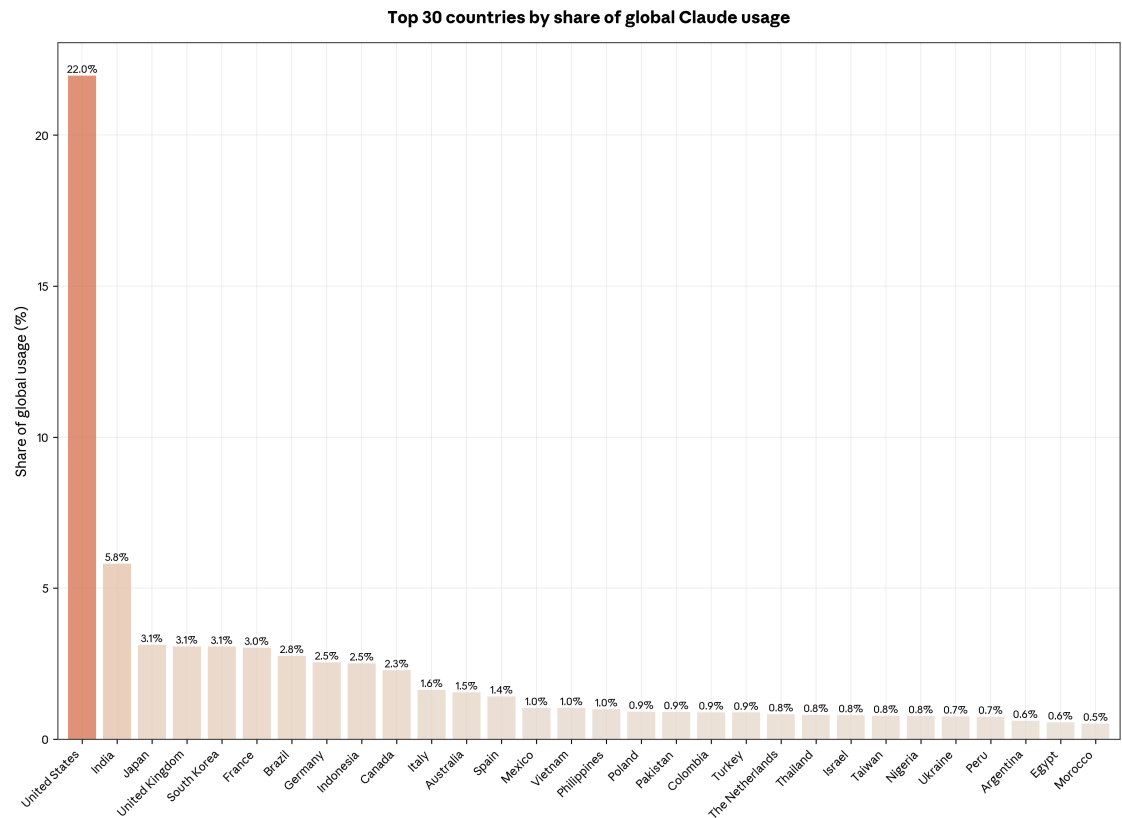


Figure 2: **Leading countries in terms of global Claude.ai usage share.** The data includes Claude.ai Free, Pro and Max conversations.

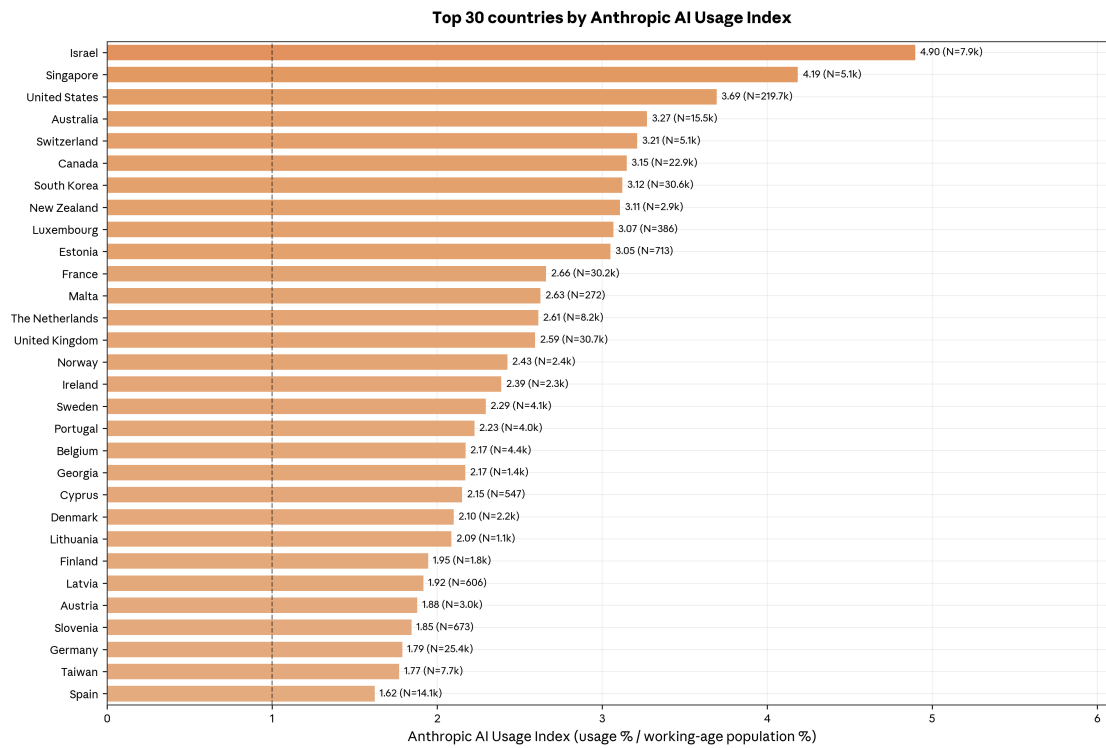


Figure 3: **Leading countries in terms of Claude.ai adoption per capita.** The figure shows the top 30 countries based on the Anthropic AI Usage Index. We only include countries with at least 200 observations in our sample for this figure because of the uncertainty of the measure for low-usage countries in our random sample. The underlying data includes Claude.ai Free, Pro and Max usage.

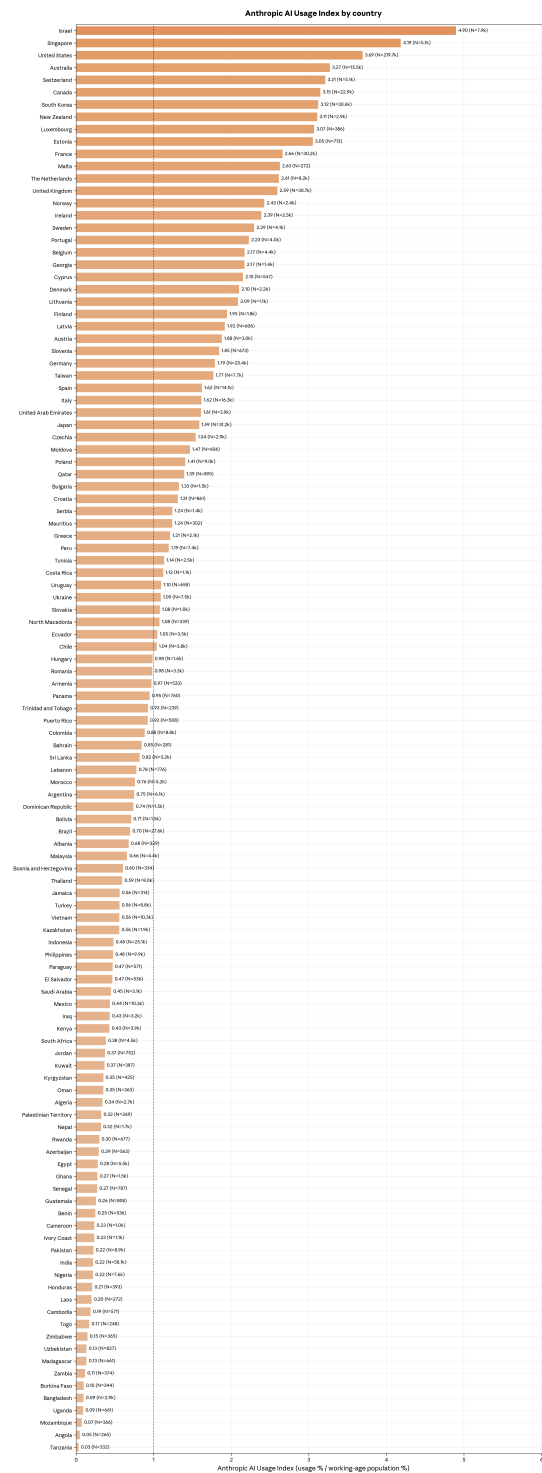


Figure 4: **Countries by Claude.ai adoption per capita.** The figure shows Anthropic AI Usage Index for all countries. We only include countries with at least 200 observations in our sample for this figure because of the uncertainty of the measure for low-usage countries in our random sample. The underlying data includes Claude.ai Free, Pro and Max usage.

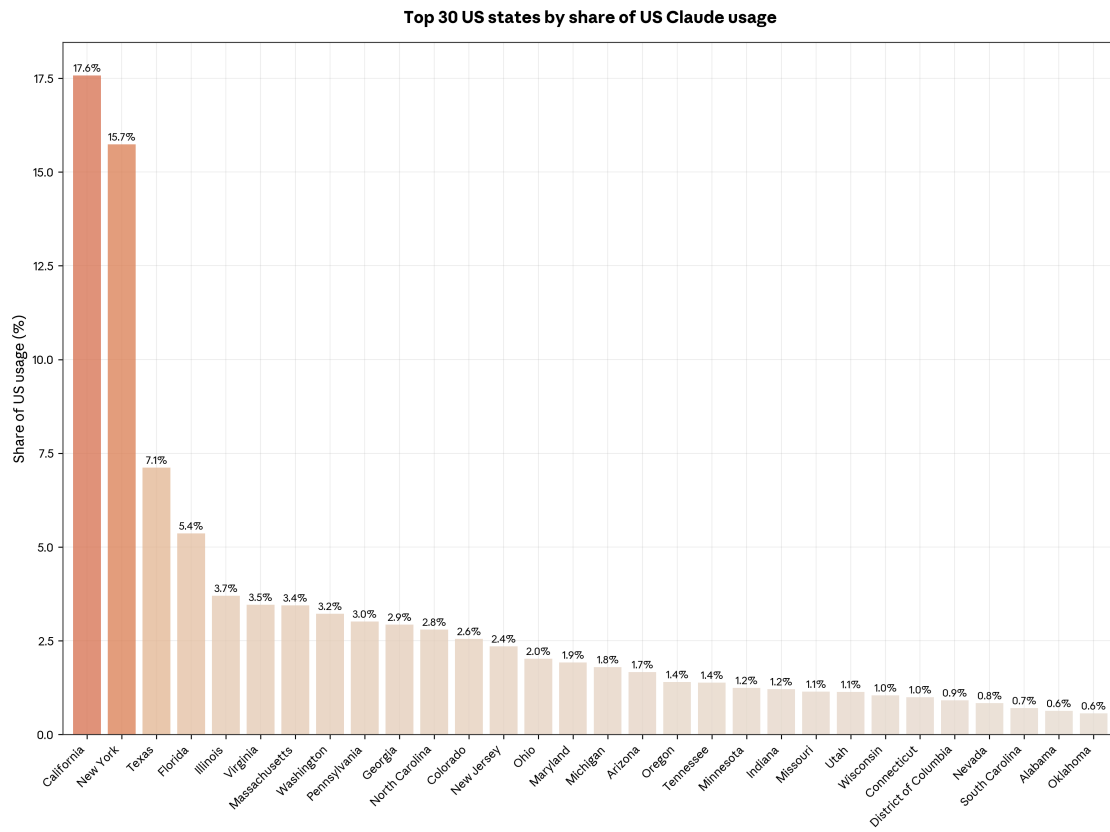


Figure 5: **Leading US states in terms of Claude.ai usage share.** The data includes Claude.ai Free, Pro and Max conversations.

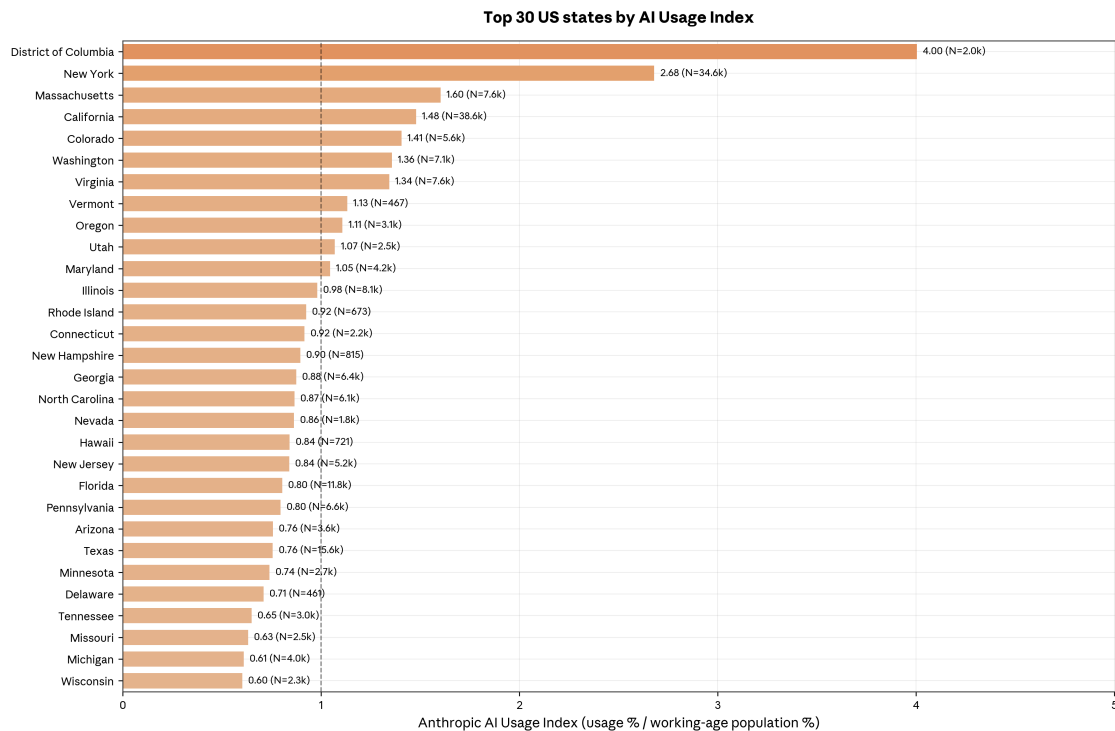


Figure 6: **Leading US states by Claude.ai adoption per capita.** The figure shows the top 30 US states based on the Anthropic AI Usage Index. We only include states with at least 100 observations in our sample for this figure because of the uncertainty of the measure for low-usage states in our random sample. The underlying data includes Claude.ai Free, Pro and Max usage.

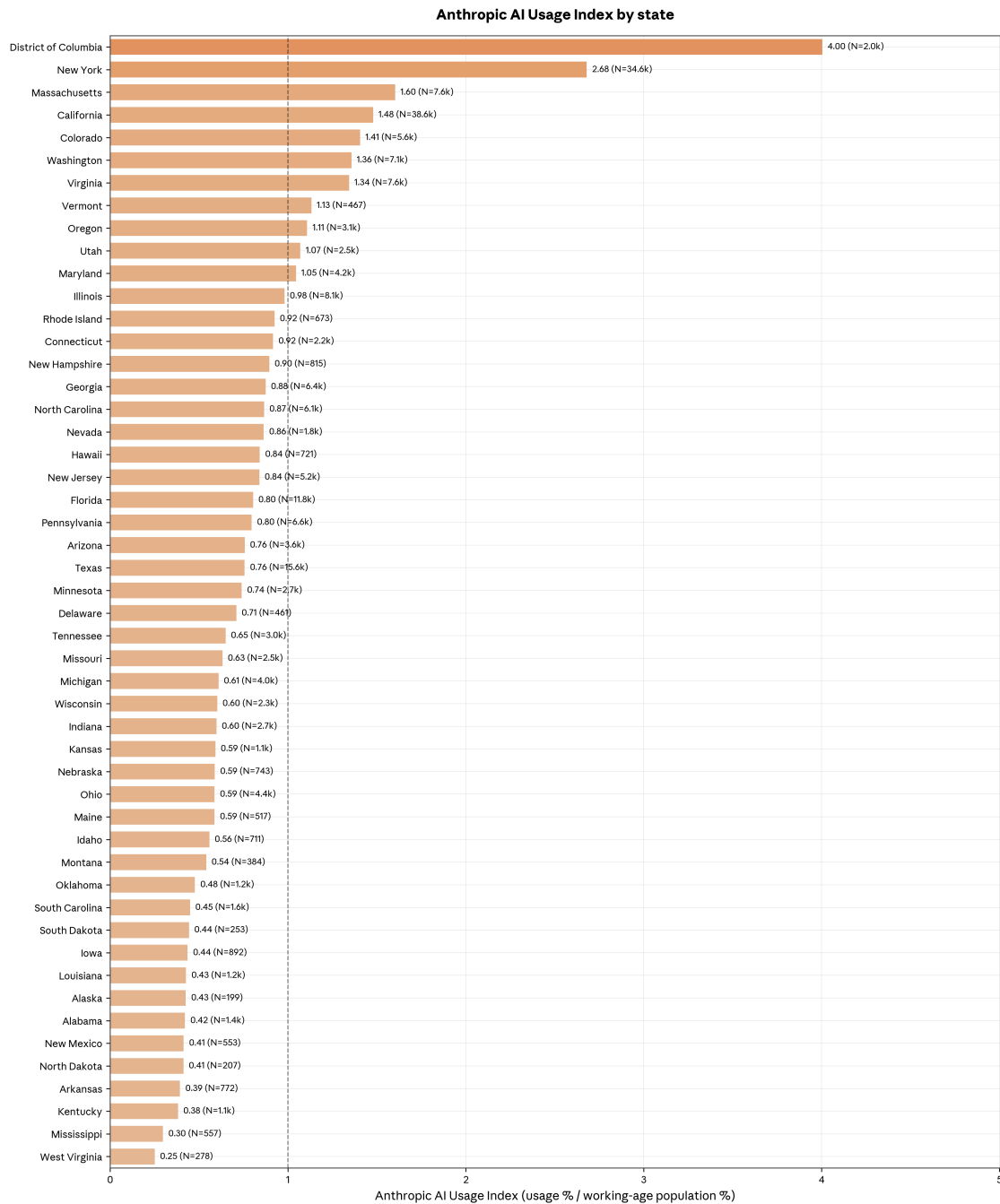


Figure 7: **US states by Claude.ai adoption per capita.** The figure shows Anthropic AI Usage Index for all US states. We only include states with at least 100 observations in our sample for this figure because of the uncertainty of the measure for low-usage states in our random sample. The underlying data includes Claude.ai Free, Pro and Max usage.

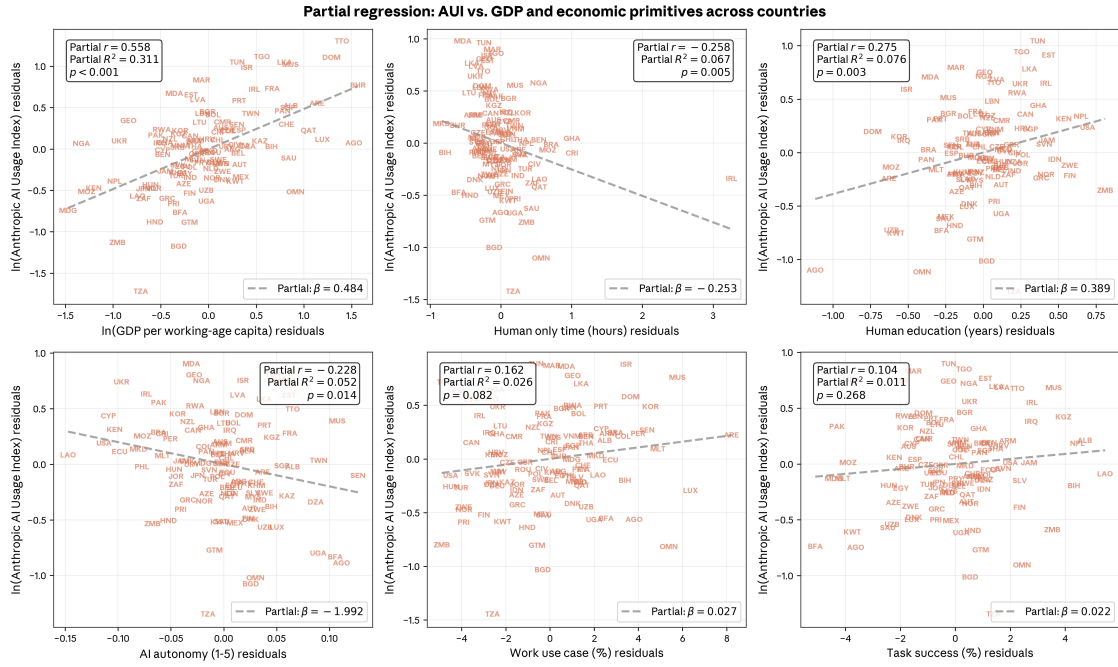


Figure 8: **Partial regression of the Anthropic AI Usage Index on five core economic primitives and GDP per capita at the country level.** Each plot shows the relationship between the natural logarithm of the Anthropic AI Usage Index and a core economic primitive as well as log GDP per capita, controlling for other core primitives. Labels show the ISO-3166-1 country codes. We only include countries with at least 200 observations in our sample for this figure because of the uncertainty of the measure for low-usage countries in our random sample. The underlying data includes Claude.ai Free, Pro and Max usage.

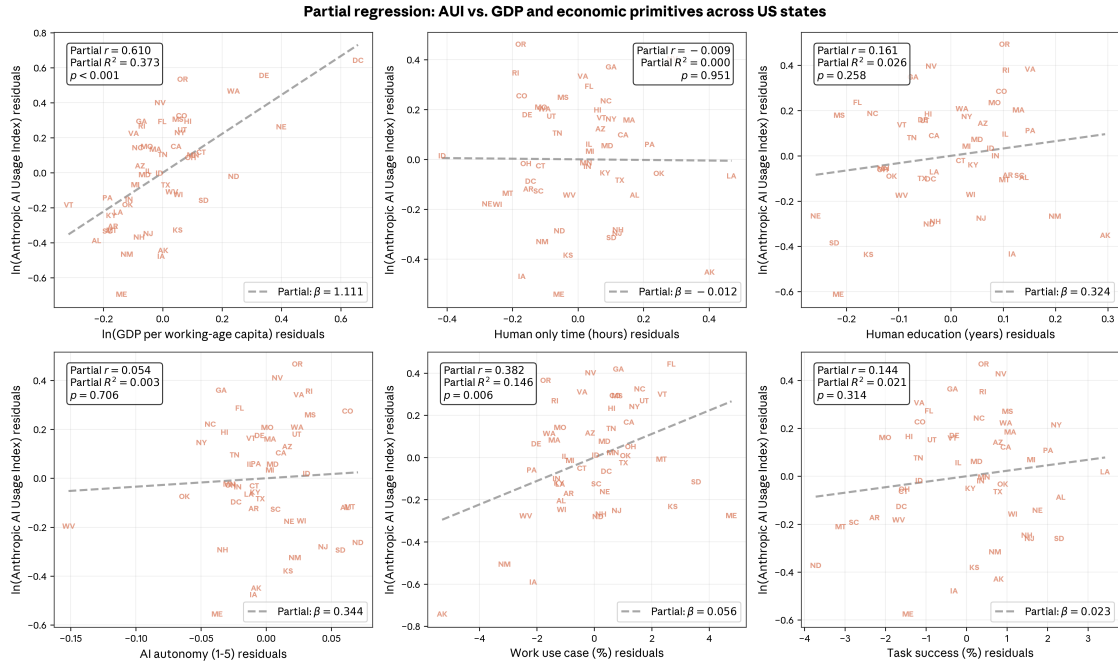


Figure 9: **Partial regression of the Anthropropic AI Usage Index on five core economic primitives and GDP per capita at the US state level.** Each plot shows the relationship between the natural logarithm of the Anthropropic AI Usage Index and a core economic primitive as well as log GDP per capita, controlling for other core primitives. Labels show the ISO-3166-2 region codes. We only include states with at least 100 observations in our sample for this figure because of the uncertainty of the measure for low-usage states in our random sample. The underlying data includes Claude.ai Free, Pro and Max usage.

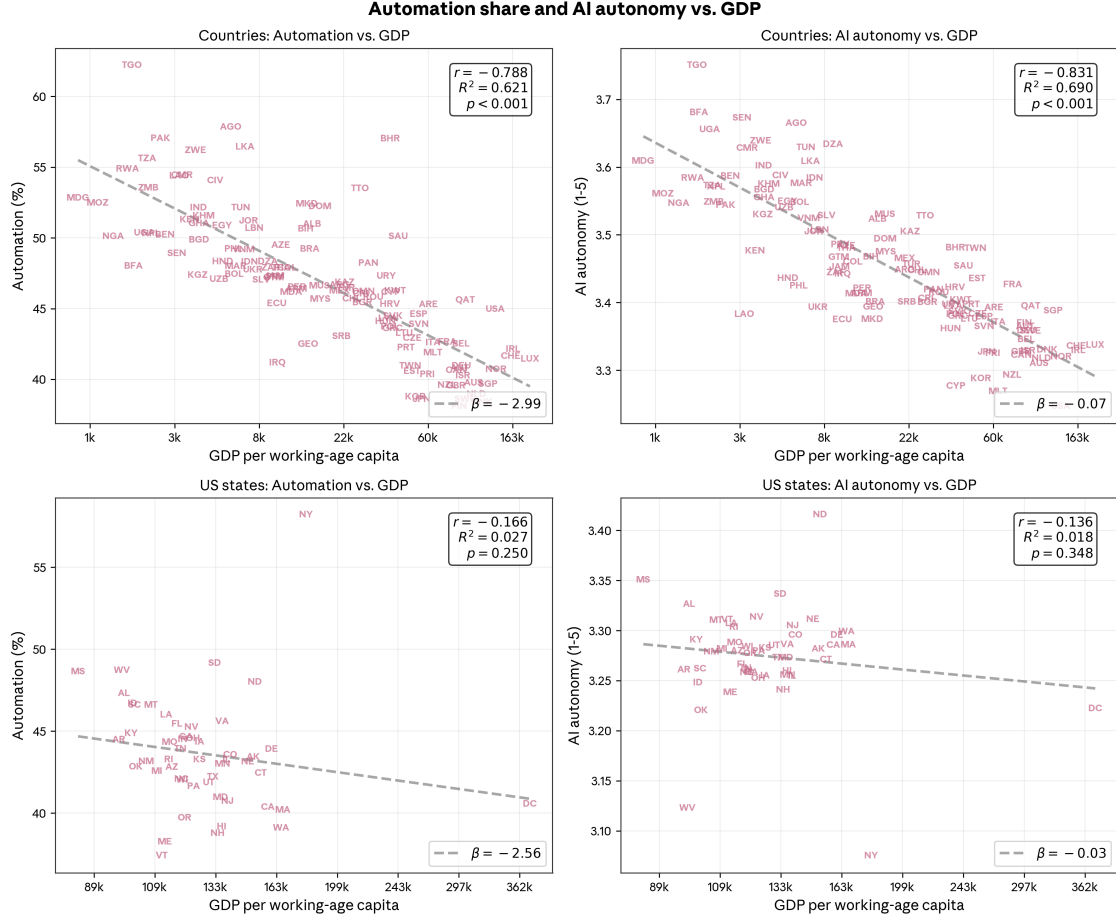


Figure 10: **Relationship between automation share or AI autonomy and GDP per working-age capita.** Each plot shows the relationship between automation share (left panels) or AI autonomy (right panels) and log GDP per working-age capita. Labels show the ISO-3166-2 region codes. We only include countries with at least 200 and states with at least 100 observations in our sample for this figure because of the uncertainty of the measure for low-usage states in our random sample. The underlying data includes Claude.ai Free, Pro and Max usage.

2 Prompts for Claude.ai and 1P API facets

2.1 Claude.ai and 1P API shared prompts

2.1.1 Multitasking

Did the User multitask in this conversation? Choose from these options:

- Yes: the User was working on multiple tasks over the course of the conversation
- No: the User was working on a single task over the course of the conversation

Your classification should be exactly one of the answer options, nothing else, provided in `<answer>` tags. For example, your answer could be `<answer>Yes</answer>` or `<answer>No</answer>`.

2.1.2 Human ability to complete task alone

Could the User have completed this task by themselves? Choose from these options:

- Yes: the User would have been able to complete the task without the Assistant, even if it would have taken more time
- No: the User would not have been able to complete the task without the Assistant, even with more time

Your classification should be exactly one of the answer options, nothing else, provided in `<answer>` tags. For example, your answer could be `<answer>Yes</answer>` or `<answer>No</answer>`.

2.1.3 Work vs. coursework vs. personal use case

Analyze whether the conversation between the User and the Assistant primarily focuses on work, coursework or personal use. Analyze the use case according to these categories:

- Work: professional use to accomplish tasks that are part of the User's job
- Coursework: use to help the User complete coursework in educational contexts
- Personal: use for any domain that is not work or coursework

Your classification should be exactly one of the answer options, nothing else. For example, your answer could be `<answer>Work</answer>` or `<answer>Coursework</answer>` or `<answer>Personal</answer>`.

2.1.4 Task Success

Did the Assistant complete the task provided by the User successfully? Choose from these options:

- Yes: the Assistant completed the task provided by the User successfully
- No: the Assistant did not complete the task provided by the User successfully

Your classification should be exactly one of the answer options, nothing else, provided in `<answer>` tags. For example, your answer could be `<answer>Yes</answer>` or `<answer>No</answer>`.

2.1.5 Human education level

Estimate how many years of formal education someone would need to understand the User prompts in this conversation. Your answer should be a single number out of the discrete numbers ranging from 0-20.

Your classification should be exactly one of the answer options, nothing else, provided in `<answer>` tags. For example, your answer could be `<answer>0</answer>` or `<answer>20</answer>`.

2.1.6 AI education level

Estimate how many years of formal education someone would need to understand the Assistant responses in this conversation. Your answer should be a single number out of the discrete numbers ranging from 0-20.

Your classification should be exactly one of the answer options, nothing else, provided in `<answer>` tags. For example, your answer could be `<answer>0</answer>` or `<answer>20</answer>`.

2.2 Claude.ai prompts

2.2.1 AI Autonomy

Human: Consider the following conversation:

```
<conversation>
{TRANSCRIPT}
</conversation>
```

Estimate how much autonomy the Assistant had to make decisions in this conversation (a discrete number ranging from 1-5, where 1 is none and 5 is extreme).

Before providing your final answer, use <thinking> tags to break down your reasoning process:

<thinking>
2-5 sentences of reasoning estimating how much autonomy the Assistant had to make decisions in this conversation.
</thinking>

Provide your output in the following format:

<answer>A discrete number representing the level of autonomy the Assistant had to make decisions (ranging from 1-5, where 1 is none and 5 is extreme)</answer>

Assistant: <thinking>

2.2.2 Human-Only Time Estimate

Human: Consider the following conversation:

<conversation>
{TRANSCRIPT}
</conversation>

Estimate how many hours a competent professional would need to complete the tasks done by the Assistant.

Assume they have:

- The necessary domain knowledge and skills
- All relevant context and background information
- Access to required tools and resources
- No access to AI tools to assist with the work

Before providing your final answer, use <thinking> tags to break down your reasoning process:

<thinking>
2-5 sentences of reasoning estimating how many hours would be needed to complete the tasks.
</thinking>

Provide your output in the following format:

<answer>A number representing hours (can use decimals like 0.5 for shorter tasks)</answer>

Assistant: <thinking>

2.2.3 Human with AI Time Estimate

Human: Consider the following conversation:

```
<conversation>
{TRANSCRIPT}
</conversation>
```

Estimate how many minutes the User spent completing the tasks in the prompt with the Assistant.

Consider:

- Number and complexity of User messages
- Time reading Assistant's responses
- Time thinking and formulating questions
- Time reviewing outputs and iterating
- Realistic typing/reading speeds
- Time implementing suggestions or running code outside of the conversation (only if directly relevant to the tasks)

Before providing your final answer, use <thinking> tags to break down your reasoning process:

```
<thinking>
2-5 sentences of reasoning about how many minutes the User spent.
</thinking>
```

Provide your output in the following format:

```
<answer>A number representing minutes</answer>
```

Assistant: <thinking>

2.2.4 Collaboration Pattern

Human: Consider the following conversation:

```
<conversation>
{TRANSCRIPT}
</conversation>
```

Your task is to analyze human-AI interactions in conversation transcripts to identify the primary collaboration pattern. Focus on how the human structures their requests and engages with the AI assistant. Analyze the interaction according to these collaboration patterns:

- Directive - Human delegates complete task execution to AI with minimal interaction
- Feedback Loop - Human and AI engage in iterative dialogue to complete task with human mainly providing feedback from the environment
- Task Iteration - Human and AI engage in iterative dialogue to complete a task with the human refining the AI outputs

- Learning - Human seeks understanding and explanation rather than direct task completion
- Validation - Human uses AI to check or validate their own work

Based on your analysis, identify which of the above is most representative of the user's response. If multiple patterns are present, select the one that is most appears most frequently. If you are unsure or there is not enough context to determine the most representative pattern, return 'None' as your answer. Use 'None' liberally---for only some conversations will this task be possible.

Before providing your final analysis for each response, use the <thinking> tags to break down your reasoning process. Consider the context, the user's words, and how they align with the defined patterns. Provide your output in the following format:

<thinking>

Your 2-3 sentence thinking about your decision. Include specific examples from the conversation and mention any relevant user response patterns observed.

</thinking>

<answer>

Your classification from the list (including only the name of the collaboration pattern) or 'None'

</answer>

Using this information, identify the primary collaboration pattern in the provided conversation. Ensure your classification is the exact name of a collaboration pattern above or 'None', without any other commentary. Put the classification in <answer> tags.

For example:

<answer>Directive</answer> or <answer>Feedback Loop</answer> or <answer>Task Iteration</answer> or <answer>Learning</answer> or <answer>Validation</answer> or <answer>None</answer>

Assistant: <thinking>

2.3 1P API prompts

2.3.1 AI Autonomy

Human: Consider the following API call.

```
<api_call>
{TRANSCRIPT}
</api_call>
```

Estimate how much autonomy the Assistant had to make decisions in this API call transcript (a discrete number ranging from 1-5, where 1 is none and 5 is extreme).

IMPORTANT: These interactions are happening via an LLM API, not an actual User-Assistant conversation. We are interested in the dynamics between the human User and the Assistant.

Before providing your final answer, use <thinking> tags to break down your reasoning process:

<thinking>

2-5 sentences of reasoning estimating how much autonomy the Assistant had to make decisions in this conversation.

</thinking>

Provide your output in the following format:

<answer>A discrete number representing the level of autonomy the Assistant had to make decisions (ranging from 1-5, where 1 is none and 5 is extreme)</answer>

Assistant: <thinking>

2.3.2 Human-Only Time Estimate

Human: Consider the following API call.

```
<api_call>
{TRANSCRIPT}
</api_call>
```

Estimate how many hours a competent professional would need to complete the tasks done by the Assistant.

Assume they have:

- The necessary domain knowledge and skills
- All relevant context and background information
- Access to required tools and resources
- No access to AI tools to assist with the work

IMPORTANT: These interactions are happening via an LLM API, not an actual User-Assistant conversation.

Before providing your final answer, use <thinking> tags to break down your reasoning process:

<thinking>

2-5 sentences of reasoning estimating how many hours would be needed to complete the tasks.

</thinking>

Provide your output in the following format:

<answer>A number representing hours (can use decimals like 0.5 for shorter tasks)</answer>

Assistant: <thinking>

2.3.3 Human with AI Time Estimate

Human: Consider the following API call.

```
<api_call>
{TRANSCRIPT}
</api_call>
```

Estimate how many minutes the User spent completing the tasks in the prompt with the Assistant.

Consider:

- Number and complexity of User messages
- Time reading Assistant's responses
- Time thinking and formulating questions
- Time reviewing outputs and iterating
- Realistic typing/reading speeds
- Time implementing suggestions or running code outside of the conversation (only if directly relevant to the tasks)

IMPORTANT: These interactions are happening via an LLM API, not an actual User-Assistant conversation.

Before providing your final answer, use <thinking> tags to break down your reasoning process:

```
<thinking>
```

2-5 sentences of reasoning about how many minutes the User spent.

```
</thinking>
```

Provide your output in the following format:

```
<answer>A number representing minutes</answer>
```

Assistant: <thinking>

2.3.4 Collaboration Pattern

Human: Consider the following API call.

```
<api_call>
{TRANSCRIPT}
</api_call>
```

Your task is to analyze human-AI interactions in API call transcripts to identify the primary collaboration pattern.

IMPORTANT: These interactions are happening via an LLM API, not an actual human-AI assistant conversation. Consider what is implied by the context in which the API is being used. We're interested in how an end user is interacting with the AI-driven feature or service, NOT how the developer is prompting the API. This could be confusing given the structure of the API oscillates between 'human' and 'assistant'. For the purposes of this task, when we say 'human', we mean 'end user'.

Analyze the interaction according to these collaboration patterns:

- Directive - Human delegates complete task execution to AI with minimal interaction
- Feedback Loop - Human and AI engage in iterative dialogue to complete task with human mainly providing feedback from the environment
- Task Iteration - Human and AI engage in iterative dialogue to complete a task with the human refining the AI outputs
- Learning - Human seeks understanding and explanation rather than direct task completion
- Validation - Human uses AI to check or validate their own work

Based on your analysis, identify which of the above is most representative of the user's response. If multiple patterns are present, select the one that is most appears most frequently. If you are unsure or there is not enough context to determine the most representative pattern, return 'None' as your answer. Use 'None' liberally---for only some conversations will this task be possible.

Before providing your final analysis for each response, use the <thinking> tags to break down your reasoning process. Consider the context, the user's words, and how they align with the defined patterns. Provide your output in the following format:

<thinking>

Your 2-3 sentence thinking about your decision. Include specific examples from the conversation and mention any relevant user response patterns observed.

</thinking>

<answer>

Your classification from the list (including only the name of the collaboration pattern) or 'None'

</answer>

Using this information, identify the primary collaboration pattern in the provided conversation. Ensure your classification is the exact name of a collaboration pattern above or 'None', without any other commentary. Put the classification in <answer> tags.

For example:

<answer>Directive</answer> or <answer>Feedback Loop</answer> or <answer>Task Iteration</answer> or <answer>Learning</answer> or <answer>Validation</answer> or <answer>None</answer>

Assistant: <thinking>